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Gaussian Sum Shifted Rayleigh Filter for Underwater Bearings-Only Target Tracking Problems

Rahul Radhakrishnan , Shovan Bhaumik , and Nutan Kumar Tomar 

Abstract—In this paper, a new Gaussian sum (GS) filter with shifted Rayleigh filter (SRF) as proposal has been developed for the bearings-only tracking (BOT) problem. The proposed filter estimates the prior and posterior probability density functions as weighted sum of several Gaussian densities, where the individual Gaussian densities are generated from the SRF. Performance of the proposed filter on the BOT problem is compared with the SRF, the GS sparse-grid Gauss–Hermite filter, and other available GS and quadrature-based filters. Performance of filters is compared in terms of a root-mean-square error (RMSE), track divergence, and computational time. The effect of initial uncertainty, measurement noise covariance, and sampling time on filtering accuracy are also studied. Finally, RMSEs of all the filters are evaluated in comparison with the Cramer–Rao lower bound. From simulation results, it is observed that the performance of the proposed GS-SRF is superior to all other nonlinear filters considered.

Index Terms—Bearings-only tracking (BOT), nonlinear filtering, state estimation.

I. INTRODUCTION

THIS paper presents a typical passive bearings-only tracking (BOT) scenario where the kinematics of a constant velocity target is estimated by a moving observer. The noise corrupted angle-only measurements are processed by the observer for obtaining the accurate estimates of the target. In defense applications, this scenario provides an advantage that the identity of the observer is not revealed. This kind of nonlinear filtering problem finds its application in many defense and civilian problems such as underwater tracking [1], aircraft surveillance [2], etc. Since the objective of the problem solution is to find out the position and velocity of a moving target, the problem is often mentioned as target motion analysis (TMA) [2]. The scenario considered in this paper consists of an observer, mounted with a passive SONAR, maneuvering to obtain estimates of the target. The measurements obtained by the SONAR are processed by a nonlinear filter and its output information fed to a suitable guidance law so that the target can be engaged successfully. The study of BOT problems for multisensor and multitarget is also reported in the literature [3].

The problem of BOT has been widely discussed in the literature, especially the autonomous TMA, where the measurements are obtained by a single sensor [4]–[6]. For nonmaneuvering or constant velocity targets, the resulting target–observer dynamic model becomes unobservable if a constant velocity model is followed by the observer [7],

[8]. Hence, for such target models, an observer maneuver is required, and we focus on such type of problems in this paper.

Bayesian framework for nonlinear filtering is the most common approach used for tackling BOT problems. In this approach, the prior and posterior probability densities that are to be computed can be represented as [9]

$$p(\mathbf{x}_k | \tilde{\theta}_{k-1}) = \int p(\mathbf{x}_k | \mathbf{x}_{k-1}) p(\mathbf{x}_{k-1} | \tilde{\theta}_{k-1}) d\mathbf{x}_{k-1} \quad (1)$$

and

$$p(\mathbf{x}_k | \tilde{\theta}_k) = \frac{p(\tilde{\theta}_k | \mathbf{x}_k) p(\mathbf{x}_k | \tilde{\theta}_{k-1})}{\int p(\tilde{\theta}_k | \mathbf{x}_k) p(\mathbf{x}_k | \tilde{\theta}_{k-1}) d\mathbf{x}_k} \quad (2)$$

where $\mathbf{x}_k$ represent the states of the system, $\tilde{\theta}_k$ represents the measurements, and k denotes time steps. Here, prior and posterior probability densities are obtained using the Chapman–Kolmogorov equation and Bayes’ rule, respectively. When the process or measurement model is nonlinear and the density function considered is non-Gaussian, the above integrals become mathematically intractable. In BOT, the solution becomes intractable due to the nonlinear nature of the measurement model [10]. Hence, we have to go for suboptimal filters that make use of various approximations.

The extended Kalman filter (EKF) was the first suboptimal filter that appeared in the literature. However, due to the linearization of the nonlinear measurement model, it suffered from poor accuracy and track divergence [2]. After EKF, several nonlinear filters, such as the unscented Kalman filter (UKF) [11], the cubature Kalman filter (CKF) [12], etc., were introduced, and were implemented for solving underwater target tracking problems [10], [13]. All these filters performed with comparatively reasonable accuracy and track divergence. For achieving higher accuracy levels, the Gauss–Hermite filter (GHF) was introduced [14]. It resulted in an exponential increase in the number of support points and weights. To reduce the computational load of the GHF without compromising accuracy, the sparse-grid Gauss–Hermite filter (SGHF) was developed [15]. Both GHF and SGHF performed with better accuracy for BOT problems when compared to UKF and CKF [8], [16].

The shifted Rayleigh filter (SRF), introduced exclusively for BOT problems, is a moment matching filter that makes use of an augmented measurement framework so that an exact calculation of its posterior density can be performed. Here, the measurements are considered to be the direction cosines of the true measured bearings. It has been claimed [17] that the only approximation introduced in the filtering framework is the approximation of the posterior density by its moment matching normal distribution. It has also been proved that the modified measurement model is almost similar to the conventional bearing model

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R. Radhakrishnan and S. Bhaumik are with the Department of Electrical Engineering, Indian Institute of Technology Patna, Patna 801106, India (e-mail: rahul.pee13@iitp.ac.in; shovan.bhaumik@iitp.ac.in).

N. Kumar Tomar is with the Department of Mathematics, Indian Institute of Technology Patna, Patna 801106, India (e-mail: nktomar@iitp.ac.in).

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[18]. All these properties led to high track accuracy even for highly challenging single sensor BOT scenarios [19].

Here, we propose a variant to nonlinear filtering for solving BOT problems where the prior and posterior densities are considered non-Gaussian and approximated as weighted sum of several Gaussian densities. Gaussian sum (GS) approach can incorporate any Kalman-based nonlinear filter so that the prior and posterior probability densities can be represented using Gaussian densities. This approach was first introduced in [20], and as time progressed, many filters such as UKF, CKF, and the particle filter were incorporated in it [10], [21], [22]. Out of these, filtering accuracy of GS-CKF for the passive underwater BOT problem was studied in detail and found to be more efficient than the existing methods [10]. Since SRF and SGHF have proved to be good solutions for BOT problems, the scope of incorporating them in GS approach for BOT problems is yet to be explored, and in this paper, we intend to bridge this gap.

In this paper, GS-SRF and GS-SGHF are developed for a typical BOT problem. We also implemented GS-EKF, GS-UKF, GS-CKF, GS-GHF, and other quadrature filters for comparing the filtering performance. Performance of the filters was validated by plotting the root-mean-square error (RMSE), the percentage of track divergence, and the computational time for two different target–observer scenarios. The Cramer–Rao lower bound (CRLB) is also plotted to compare the estimation error of filters with its theoretical lower bound. Convergence study for various filters is done based on initial uncertainty, measurement noise, and sampling time, and their effect on track divergence is studied. From simulation results, it can be observed that the proposed GS-SRF performed with almost zero track divergence for the scenarios considered, which is the best among all the available filters.

II. GAUSSIAN SUM FILTERING

In the GS approach, prior and posterior densities given in (1) and (2) are represented as a weighted sum of individual Gaussian densities [10], [22]. Thus, the prior density could be expressed as

$$p(\mathbf{x}_k | \tilde{\theta}_{k-1}) \approx \sum_{i=1}^{N_g} w_{k|k-1}^i \mathcal{N}(\mathbf{x}_k; \mu_{k|k-1}^i, P_{k|k-1}^i) \quad (3)$$

where N_g is the number of Gaussian components, w^i is their corresponding weights, and $i = 1, \dots, N_g$. Here, we propose SRF and SGHF for generating the individual Gaussian densities.

The posterior density can be written as

$$p(\mathbf{x}_k | \tilde{\theta}_k) \propto p(\tilde{\theta}_k | \mathbf{x}_k) p(\mathbf{x}_k | \tilde{\theta}_{k-1}) \approx \sum_{i=1}^{N_g} w_{k|k}^i \mathcal{N}(\mathbf{x}_k; \mu_{k|k}^i, P_{k|k}^i) \quad (4)$$

where the update of weights in the presence of measurements is done using

$$w_{k|k}^i = \frac{p(\tilde{\theta}_k | \mathbf{x}_k, i) w_{k|k-1}^i}{\sum_{i=1}^{N_g} p(\tilde{\theta}_k | \mathbf{x}_k, i) w_{k|k-1}^i}. \quad (5)$$

The posterior means $\mu_{k|k}^i$ and covariances $P_{k|k}^i$ of the i th Gaussian component can be determined with the help of nonlinear filters. From the weight equation mentioned above, it is clear that the weights of Gaussian components are updated during the measurement update step, whereas during prediction, earlier step's weights are being used.

III. GS-SRF FOR BOT PROBLEM

The SRF has proved to be a highly accurate and computationally efficient nonlinear filter for solving BOT problems. This is achieved by reformulating the conventional measurement equation in terms of an augmented measurement, which is the relative position vector between a target and an observer. Given a Gaussian prior, we find the exact mean and covariance of the posterior density. At the end of each Kalman filtering cycle, this exact mean and covariance are used to approximate the non-Gaussian posterior as a Gaussian density. It has been shown that the bearing density of the modified measurement equation, which is essential for the formulation of GS-SRF, is similar to the Gaussian density of the conventional measurement model [17].

Let us consider a stochastic linear process model for target dynamics expressed as [2], [8]

$$\mathbf{x}_k = F \mathbf{x}_{k-1} + v_{k-1} - \bar{v}_{k-1,k} \quad (6)$$

where $\mathbf{x}_k = [x_k \ y_k \ \dot{x}_k \ \dot{y}_k]^T$ are the relative states, F is the state transition matrix, $\bar{v}_{k-1,k}$ is defined as a vector of observer inputs, and v_{k-1} is a Gaussian process noise vector with zero mean and covariance matrix Q_k . Now, the nonlinear bearing measurement can be expressed as $\hat{\theta}_k = \theta_k + \bar{\gamma}_k$, where $\bar{\gamma}_k$ is modeled as a Gaussian noise with zero mean and standard deviation σ_{θ} . The true bearing measurement can be expressed as $\theta_k = \tan^{-1}(\dot{x}_k / \dot{y}_k)$. These are obtained in a direction from the observer to the target with a reference clockwise positive to the y-axis.

Now, an augmented measurement model, which is the relative position vector between the target and the observer, is defined as $\bar{y}_k = H \mathbf{x}_k + \gamma_k$. Here, $H = [I_{2 \times 2} \ O_{2 \times 2}]$, where $I_{2 \times 2}$ is an identity matrix of order 2 and $O_{2 \times 2}$ is a zero matrix of order 2. The actual measurement $b_k = \Pi(\bar{y}_k)$ is a vector of direction cosines to the state $\mathbf{x}_k$ corrupted with noise. It can also be interpreted as the projection of $\bar{y}_k$ on to the unit circle, that is, $b_k = \bar{y}_k / \|\bar{y}_k\| = [\sin(\hat{\theta}_k) \ \cos(\hat{\theta}_k)]^T$, where $\|\bar{y}_k\|$ is the range. Here, γ_k is a white Gaussian noise with zero mean and covariance matrix $R_k = \sigma_{\theta}^2 E[\|H \mathbf{x}_k\|^2] I_{2 \times 2}$. It can be further expanded as

$$\begin{aligned} R_k &= \sigma_{\theta}^2 E[x_{k|k-1}^2 + y_{k|k-1}^2] I_{2 \times 2} \\ &= \sigma_{\theta}^2 \{E[x_{k|k-1} - \hat{x}_{k|k-1}]^2 + \hat{x}_{k|k-1}^2 + E[y_{k|k-1} - \hat{y}_{k|k-1}]^2 \\ &\quad + \hat{y}_{k|k-1}^2\} I_{2 \times 2} \\ &= \sigma_{\theta}^2 \{\text{var}[x_{k|k-1}] + \hat{x}_{k|k-1}^2 + \text{var}[y_{k|k-1}] + \hat{y}_{k|k-1}^2\} I_{2 \times 2} \end{aligned} \quad (7)$$

where $\hat{x}_{k|k-1}^2$ and $\hat{y}_{k|k-1}^2$ can be obtained from the prior mean $\hat{\mathbf{x}}_{k|k-1}$. Similarly, $\text{var}[x_{k|k-1}]$ and $\text{var}[y_{k|k-1}]$ can be obtained from the prior covariance $P_{k|k-1}$.

SRF equations are derived assuming direct access to the augmented measurement $\bar{y}_k$. However, instead of $\bar{y}_k$, b_k is the only available measurement. Hence, filtering equations are modified so that the unavailability of $\bar{y}_k$ is accounted for by evaluating moments of $\mathbf{x}_k$ conditioned on b_k . Here, we choose to write the prior and posterior densities of $\mathbf{x}_k$ conditioned on $\bar{y}_k$.

To formulate the GS-SRF, the prior and posterior non-Gaussian probability densities [mentioned in (1) and (2)] are approximated using the weighted sum of several Gaussian densities, obtained from the SRF. First, we assume that the available posterior pdf is in the form [20], [23]

$$p(\mathbf{x}_{k-1} | \bar{y}_{k-1}) \approx \sum_{i=1}^{N_g} w_{k-1|k-1}^i \mathcal{N}(\mathbf{x}_{k-1}; \mu_{k-1|k-1}^i, P_{k-1|k-1}^i). \quad (8)$$

Now, the prior density given in (3) for the GS-SRF formulation can be expressed as $p(\mathbf{x}_k | \bar{y}_{k-1})$. To find this density, mean $\mu_{k|k-1}^i$ and

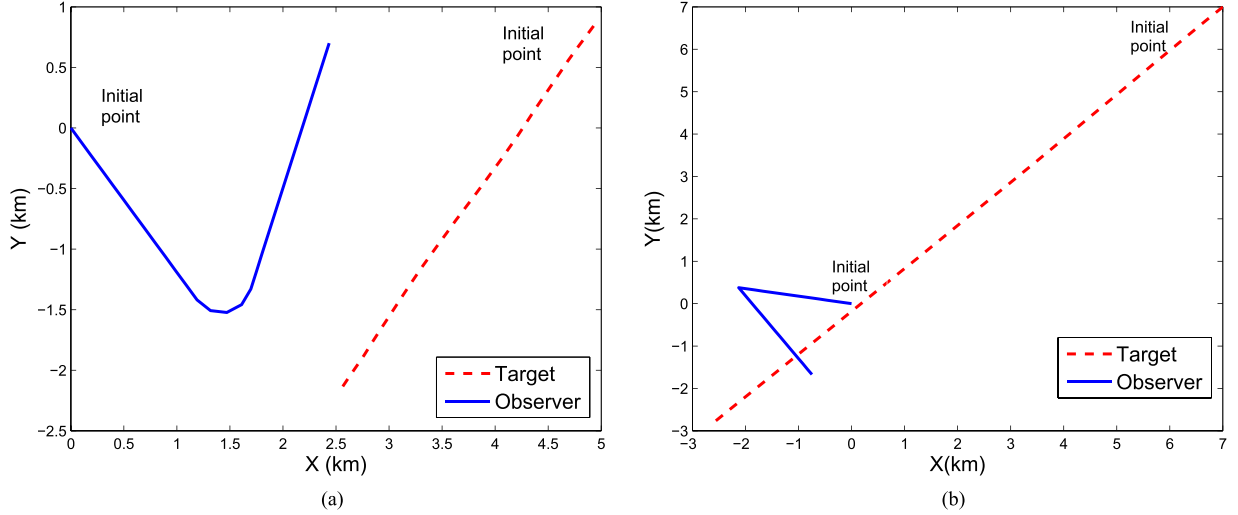


Fig. 1. Target and observer dynamics. (a) Target and observer dynamics for case 1. (b) Target and observer dynamics for case 2.

covariance $P_{k|k-1}^i$ have to be calculated. These are obtained from the SRF equations as

$$\mu_{k|k-1}^i = F \mu_{k-1|k-1}^i - \mathcal{U}_{k-1,k} \quad (9)$$

and

$$P_{k|k-1}^i = F P_{k-1|k-1}^i F^T + Q. \quad (10)$$

Similarly, the posterior density given in (4) for the GS-SRF formulation is expressed as $p(\mathbf{x}_k | \bar{\mathbf{y}}_k)$, where the posterior mean and covariance are defined as

$$\mu_{k|k}^i = \mu_{k|k-1}^i + K_k^i [\zeta_k^i b_k - H \mu_{k|k-1}^i] \quad (11)$$

and

$$P_{k|k}^i = (I - K_k^i H) P_{k|k-1}^i + \delta_k^i K_k^i b_k b_k^T K_k^{iT} \quad (12)$$

where Kalman gain $K_k^i = P_{k|k-1}^i H^T V_k^{i-1}$ and $V_k^i = H P_{k|k-1}^i H^T + R_k^i$.

Here

$$\zeta_k^i = E[\|\bar{\mathbf{y}}_k^i\| | b_k] = [b_k^T V_k^{i-1} b_k]^{-1/2} \rho(\mathbf{z}_k^i) \quad (13)$$

and

$$\delta_k^i = \text{var}[\|\bar{\mathbf{y}}_k^i\| | b_k] = [b_k^T V_k^{i-1} b_k]^{-1} (2 + \mathbf{z}_k^{iT} \rho(\mathbf{z}_k^i) - \rho(\mathbf{z}_k^i)^2) \quad (14)$$

where the variable $\mathbf{z}_k^i = [b_k^T V_k^{i-1} b_k]^{-1/2} b_k^T V_k^{i-1} [H \mu_{k|k-1}^i]$. In these equations, $\rho(\mathbf{z}_k^i)$ is the mean of a shifted Rayleigh variable [17] expressed as

$$\begin{aligned} \rho(\mathbf{z}_k^i) &= \frac{\int_0^\infty s^2 \exp^{-1/2(s-\mathbf{z}_k^i)^2} ds}{\int_0^\infty s \exp^{-1/2(s-\mathbf{z}_k^i)^2} ds} \\ &= \frac{\mathbf{z}_k^i \exp^{-\mathbf{z}_k^{i2}/2} + \sqrt{2\pi} (\mathbf{z}_k^{i2} + 1) \mathbb{F}(\mathbf{z}_k^i)}{\exp^{-\mathbf{z}_k^{i2}/2} + \sqrt{2\pi} \mathbf{z}_k^i \mathbb{F}(\mathbf{z}_k^i)} \end{aligned} \quad (15)$$

where $\mathbb{F}(\mathbf{z}_k^i)$ is the cumulative distribution function of a standard normal variable. For updating the weights of the GS filter in the presence of measurements, (5) has to be made use of, where the bearing density $p(\bar{\mathbf{y}}_k)$ has to be calculated.

A. Bearing Density

Here, we derive the bearing density which is essential for the formulation of GS-SRF. Let $\bar{\mathbf{y}}_k = [x_k \ y_k]^T$ be a Gaussian random variable with mean $m = [m_1 \ m_2]^T = H \mu_{k|k-1}$ and covariance V . Then, its distribution in polar coordinates can be expressed as

$$p(r, \theta) = \frac{r}{2\pi \sqrt{|V|}} \exp^{-1/2(g(r, \theta) - m)^T V^{-1} (g(r, \theta) - m)} \quad (16)$$

where $g(r, \theta) = [r \cos(\theta) \ r \sin(\theta)]^T$. This can be simplified as

$$p(r, \theta) = \frac{r}{2\pi \sqrt{|V|}} \exp^{-1/2(Ar^2 + Br + C)} \quad (17)$$

where

$$\begin{aligned} A &= a_{11} \sin^2(\theta) + (a_{12} + a_{21}) \cos(\theta) \sin(\theta) + a_{22} \cos^2(\theta) \\ B &= -2a_{11} m_1 \sin(\theta) - 2a_{22} m_2 \cos(\theta) \\ &\quad - (a_{12} + a_{21})(m_1 \cos(\theta) + m_2 \sin(\theta)) \\ C &= a_{11} m_1^2 + a_{21} m_1 m_2 + a_{12} m_1 m_2 + a_{22} m_2^2 \end{aligned} \quad (18)$$

and a_{ij} are the elements of V^{-1} .

Density of θ can be found by integrating out r . Rearranging $p(r, \theta)$ as

$$p(r, \theta) = \frac{r}{2\pi \sqrt{|V|}} \exp^{\frac{B^2}{8A} - \frac{C}{2}} \exp^{-\frac{A}{2}(r + \frac{B}{2A})^2} \quad (19)$$

we have

$$\begin{aligned} p(\theta) &= \frac{1}{2\pi \sqrt{|V|}} \exp^{\frac{B^2}{8A} - \frac{C}{2}} \int_0^\infty r \exp^{-\frac{A}{2}(r + \frac{B}{2A})^2} dr \\ &= \frac{\exp^{\frac{B^2}{8A} - \frac{C}{2}}}{2\pi \sqrt{|V|}} \left(\frac{1}{A} \exp^{-\frac{B^2}{8A}} - \frac{\sqrt{2\pi} B}{2A^{3/2}} \mathbb{F}\left(-\frac{B}{2\sqrt{A}}\right) \right). \end{aligned} \quad (20)$$

Similar formulation can be done for developing GS-SGHF. For more information on SGHF, refer to [15].

IV. PROBLEM FORMULATION

In this paper, we consider two tracking scenarios: case 1 and case 2, as shown in Fig. 1. From the figure, it can be observed that the target is following a constant velocity model and the observer, a maneuvered

TABLE I
TRACKING SCENARIO PARAMETERS

Parameters	Case 1	Case 2
Initial range (r)	5 km	10 km
Target speed (s)	4 kn	15 kn
Target course	-140°	-135.4°
Observer speed	5 kn	5 kn
Observer initial course	140°	-80°
Observer final course	20°	146°
Observer maneuver	From 13th to 17th min	15th min
σ_θ	1.5°	2°
σ_c	$\pi/\sqrt{12}$	$\pi/\sqrt{12}$
$\tilde{q}$	$1.944 \times 10^{-6} \text{ km}^2/\text{min}^3$	$2.142 \times 10^{-6} \text{ km}^2/\text{min}^3$

path which is modeled in different phases. The motive of this paper is to estimate the position and velocity of the target, defined as $\mathbf{x}_k^t = [x_k^t \ y_k^t \ \dot{x}_k^t \ \dot{y}_k^t]^T$. Similarly, observer state variables can be defined as $\mathbf{x}_k^o = [x_k^o \ y_k^o \ \dot{x}_k^o \ \dot{y}_k^o]^T$. Now, a relative state vector can be defined as $\mathbf{x}_k \triangleq \mathbf{x}_k^t - \mathbf{x}_k^o$.

The process model (6) is considered where F is the state transition matrix given as

$$F = \begin{bmatrix} \mathbf{I}_{2 \times 2} & T\mathbf{I}_{2 \times 2} \\ \mathbf{O}_{2 \times 2} & \mathbf{I}_{2 \times 2} \end{bmatrix}$$

and covariance matrix Q of process noise v_{k-1} is

$$Q = \begin{bmatrix} \frac{T^3}{3} \mathbf{I}_{2 \times 2} & \frac{T^2}{2} \mathbf{I}_{2 \times 2} \\ \frac{T^2}{2} \mathbf{I}_{2 \times 2} & T\mathbf{I}_{2 \times 2} \end{bmatrix} \tilde{q}$$

where T is the sampling interval and $\tilde{q}$ is the process noise intensity whose numerical value is taken as per [8] and [10], and mentioned in Table I. To account for the observer maneuver in the target dynamics, $\mathbf{U}_{k-1,k}$ is defined, which is a vector of observer inputs given by

$$\mathbf{U}_{k-1,k} = \begin{bmatrix} x_k^o - x_{k-1}^o - T\dot{x}_{k-1}^o \\ y_k^o - y_{k-1}^o - T\dot{y}_{k-1}^o \\ \dot{x}_k^o - \dot{x}_{k-1}^o \\ \dot{y}_k^o - \dot{y}_{k-1}^o \end{bmatrix}.$$

These observer values are usually obtained by an onboard inertial navigation system powered by a global positioning system.

In case 1, we consider a moderately nonlinear scenario [10], as shown in Fig. 1(a). The parameters required for generating the observer and target trajectory are defined in Table I, where the process model is the same as defined in (6). Case 2 considers a highly nonlinear scenario where there is an abrupt change in the bearing, which makes it more challenging for the filters to track the target. This scenario is presented in Fig. 1(b), with all the relevant parameters given in Table I.

V. SIMULATION RESULTS

In this section, performance of the proposed GS-SRF and GS-SGHF is compared with the performance of CKF, GHF, UKF, EKF, and with their GS counterparts out of 10 000 independent Monte Carlo runs. For each run, the truth target states and the filters are implemented with MATLAB software. The parameters considered for simulating the two tracking scenarios are mentioned in Table I.

A. Filter Initialization

The filters are all initialized using the method given in [2], [8], and [10]. To initialize our filters, we need to provide initial assumption on the bearing angle, the initial range, and the target speed. Filters' bearing value is initialized with the first bearing measurement $\tilde{\theta}_0$. Initial range r and initial speed s of the target are purely designer's guess. Depending on how the target is closing in on the observer, the initial course estimate can be defined as $\bar{c} = \tilde{\theta}_0 + \pi$. Finally, the initial state vector $\hat{\mathbf{x}}_{0|0}$ is calculated as

$$\hat{\mathbf{x}}_{0|0} = \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{\dot{x}} \\ \hat{\dot{y}} \end{bmatrix} = \begin{bmatrix} r \sin(\tilde{\theta}_0) \\ r \cos(\tilde{\theta}_0) \\ s \sin(\bar{c}) - \dot{x}_0^o \\ s \cos(\bar{c}) - \dot{y}_0^o \end{bmatrix} \quad (21)$$

and the initial covariance is defined as

$$P_{0|0} = \begin{bmatrix} P_{xx} & P_{xy} & 0 & 0 \\ P_{yx} & P_{yy} & 0 & 0 \\ 0 & 0 & P_{\dot{x}\dot{x}} & P_{\dot{x}\dot{y}} \\ 0 & 0 & P_{\dot{y}\dot{x}} & P_{\dot{y}\dot{y}} \end{bmatrix} \quad (22)$$

where $P_{xx} = r^2 \sigma_\theta^2 \cos^2(\tilde{\theta}_0) + \sigma_r^2 \sin^2(\tilde{\theta}_0)$, $P_{yy} = r^2 \sigma_\theta^2 \sin^2(\tilde{\theta}_0) + \sigma_r^2 \cos^2(\tilde{\theta}_0)$, $P_{xy} = P_{yx} = (\sigma_r^2 - r^2 \sigma_\theta^2) \sin \tilde{\theta}_0 \cos \tilde{\theta}_0$, $P_{\dot{x}\dot{x}} = s^2 \sigma_c^2 \cos^2 \bar{c} + \sigma_s^2 \sin^2 \bar{c}$, $P_{\dot{y}\dot{y}} = s^2 \sigma_c^2 \sin^2 \bar{c} + \sigma_s^2 \cos^2 \bar{c}$, and $P_{\dot{x}\dot{y}} = P_{\dot{y}\dot{x}} = (\sigma_s^2 - s^2 \sigma_c^2) \sin \bar{c} \cos \bar{c}$.

B. Performance Criteria

Performance analysis of various filtering techniques was done by obtaining RMSE, track divergence, and the computational time. The effect of initial uncertainty, measurement noise, and sampling time on filter accuracy is also studied. The performance criteria are briefly discussed in the following.

- 1) Root-mean-square error: RMSE in resultant position and velocity for various filters can be found using

$$\text{RMSE}_k = \sqrt{\frac{1}{M} \sum_{j=1}^M \left[\left(x_{j,k}^t - \hat{x}_{j,k}^t \right)^2 + \left(y_{j,k}^t - \hat{y}_{j,k}^t \right)^2 \right]}$$

where k denotes the total number of time steps and M is the total number of Monte Carlo runs.

- 2) Track divergence: A track is considered divergent when the estimate moves away from the truth path without converging. So we consider track divergence in such conditions where the difference between the true position and estimated position is more than a certain bound, denoted as T_b . The value of T_b is set according to the value of standard deviation of the computed position error distribution and is mentioned in the simulation section. Hence, track divergence is defined as $\text{pos}_{\text{error}} < T_b$, where $\text{pos}_{\text{error}} = \sqrt{(x_{k\text{max}}^t - \hat{x}_{k\text{max}}^t)^2 + (y_{k\text{max}}^t - \hat{y}_{k\text{max}}^t)^2}$ and $k\text{max}$ is the final time step.
- 3) Normalized (state) estimation error squared (NEES): This criterion is used to check the consistency of the filters as mentioned in [24]. NEES is defined as [10]

$$\mathbf{N}_k = (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}) P_{k|k}^{-1} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k})^T. \quad (23)$$

Under the hypothesis that the system is consistent and approximately linear Gaussian, $\mathbf{N}_k$ is chi-square distributed with n degrees of freedom, where n is the order of the system. That is, $E[\mathbf{N}_k] = n$. A more realistic measure of filter consistency is

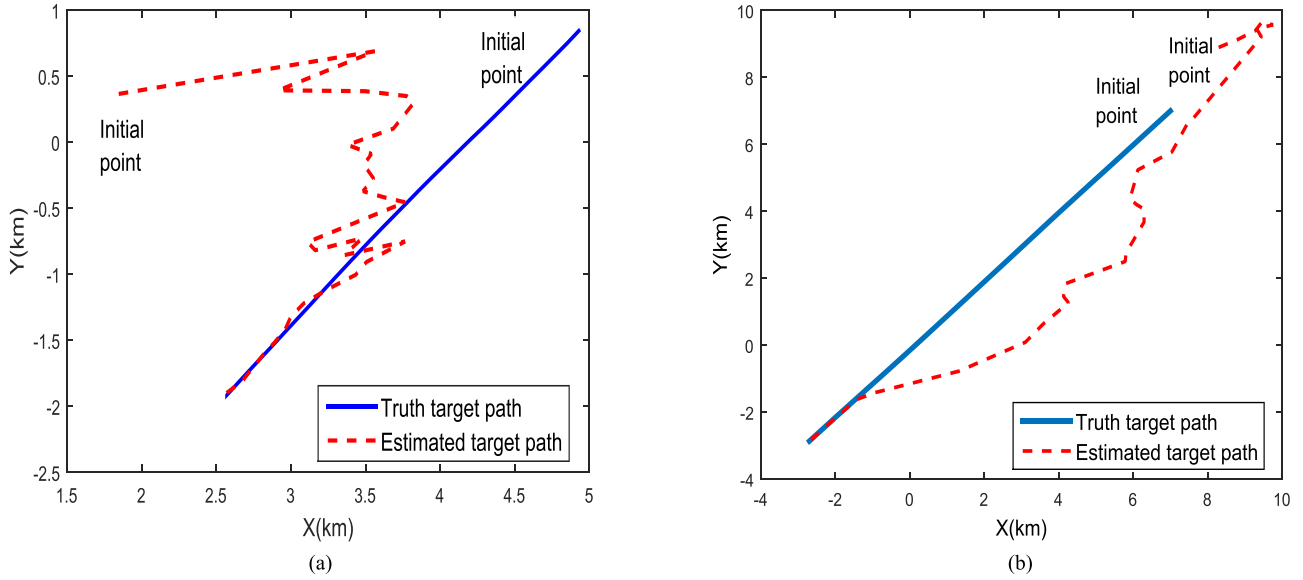


Fig. 2. Truth and estimated target path given by GS-SRF. (a) Truth and estimated target path for case 1. (b) Truth and estimated target path for case 2.

found by calculating average NEES over ensemble. This is because an inconsistent filter may appear consistent and *vice versa* for a single run [25]. Hence, average NEES can be defined as

$$\bar{N}_k = \frac{1}{M} \sum_{j=1}^M N_k^j. \quad (24)$$

Under the same assumptions as above, $M\bar{N}_k$ will have a chi-square distribution with Mn degrees of freedom. Now the consistency test is provided by defining a probability concentration region in which $\bar{N}_k \in [r_l, r_u]$, where the concentration region is calculated according to the relation

$$P\{\bar{N}_k \in [r_l, r_u]\} = 1 - \alpha \quad (25)$$

where α is the significance value such that $0 < \alpha < 1$.

C. Performance Comparison

For simulation, sampling time was taken as $T = 1$ min and the total observation period lasted for 30 min. Accuracy level for SGHF and GS-SGHF is taken as 2 and the univariate points considered for GHF and GS-GHF are 3. The displacement parameter used for splitting a multivariate Gaussian distribution is taken as 0.5, where the density is split into a total of three Gaussian components [26].

Fig. 2 shows the tracking performance of GS-SRF for both the cases, where the estimated target path is plotted with truth target path for a single run. Here, Fig. 2(a) is for case 1 and Fig. 2(b) is for case 2. From these figures, it can be observed that GS-SRF accurately tracks the target.

To compare the estimation accuracy of various filters, RMSEs in position and velocity are plotted in Figs. 3 and 4 where the diverged tracks are excluded (as per our definition for track divergence). Figs. 3(a) and 4(a) are RMSEs in position and velocity for case 1 and it was observed that convergence starts only after an observer maneuver. For case 1, EKF and its GS formulation performed with less accuracy, whereas quadrature-based filters and their GS extensions performed with better and comparable accuracy. On the other hand, SRF and GS-SRF performed with better accuracy than all quadrature-based filters and their GS extensions. It was noted that GS-SRF converges much

faster with a terminal error of 119 m. The corresponding error for SRF was 122 m, with all the quadrature-based filters giving an accuracy of around 140 m. However, RMSE in position of GS-SRF is still a bit higher than CRLB.

RMSE in position and velocity after excluding the diverged tracks for case 2 is plotted in Figs. 3(b) and 4(b). Since this is a highly nonlinear scenario, it takes more time for the filters to converge after an observer maneuver. In this case also, for RMSE in position, GS-SRF performed with better accuracy, which is comparable with SRF. In case of RMSE in velocity, GS-SRF gave more accurate result than all other filters. Here, the terminal error given by GS-SRF is 288 m, while the corresponding error for SRF was 320 m. The terminal error for all the quadrature filters ranged between 500 and 700 m. Here also, the RMSE in position of GS-SRF is higher than CRLB.

In Fig. 3(b), it can be noted that the RMSE of certain filters is below the CRLB in the initial stages after an observer maneuver. RMSE lower than CRLB may occur in rare cases and has already been reported in the literature for BOT problems [27], which was termed as superefficiency [28]. The main reason for superefficiency is due to the favorable initial conditions fed to the estimators. However, this is not the case here. Instead, it is due to the exclusion of diverged tracks while calculating the RMSE. This is the reason why even after incurring more diverged tracks than SRF and GS-SRF, RMSEs in position for certain filters are lesser than CRLB in the initial stages. In our simulation, it has been observed that the inclusion of diverged tracks results in higher RMSE values than CRLB.

To further explore the tracking efficiency and robustness of GS-SRF, a parametric study was conducted where the effects of increasing initial uncertainty, measurement noise covariance, and sampling time were considered. To study the effect of initial uncertainty, we vary the initial covariance $P_{0|0}$ according to the relation $P = \mu \times P_{0|0}$, where $\mu = 1, \dots, 5$. The effect of first two parameters, that is, the initial uncertainty and measurement noise covariance, is illustrated by plotting RMSE in position. Fig. 5(a) and (b) shows the effect of varying μ in RMSE for GS-SRF in both cases 1 and 2, respectively. From the figure, it can be noted that the estimation error increases until the observer makes a maneuver and, afterward, convergence is achieved. Here also, it can be noted that there is less impact on the terminal error as it varies from 119 m ($\mu = 1$) to 175 m ($\mu = 5$) for case 1 and from 288 to 640 m

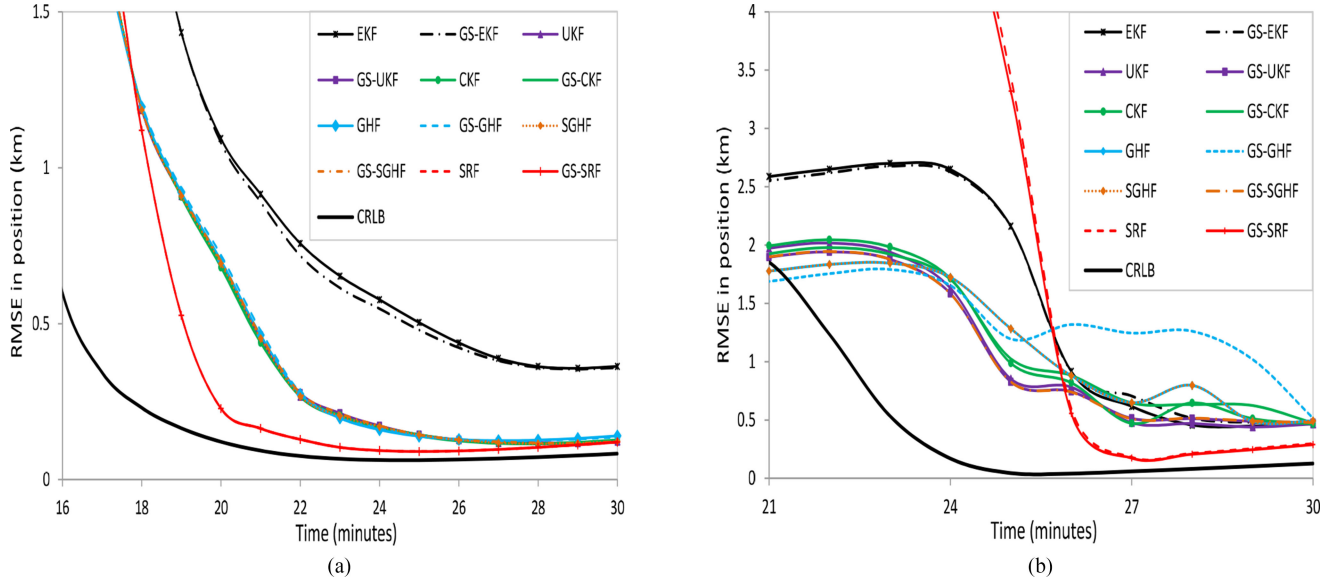


Fig. 3. RMSE in position for (a) case 1 and (b) case 2.

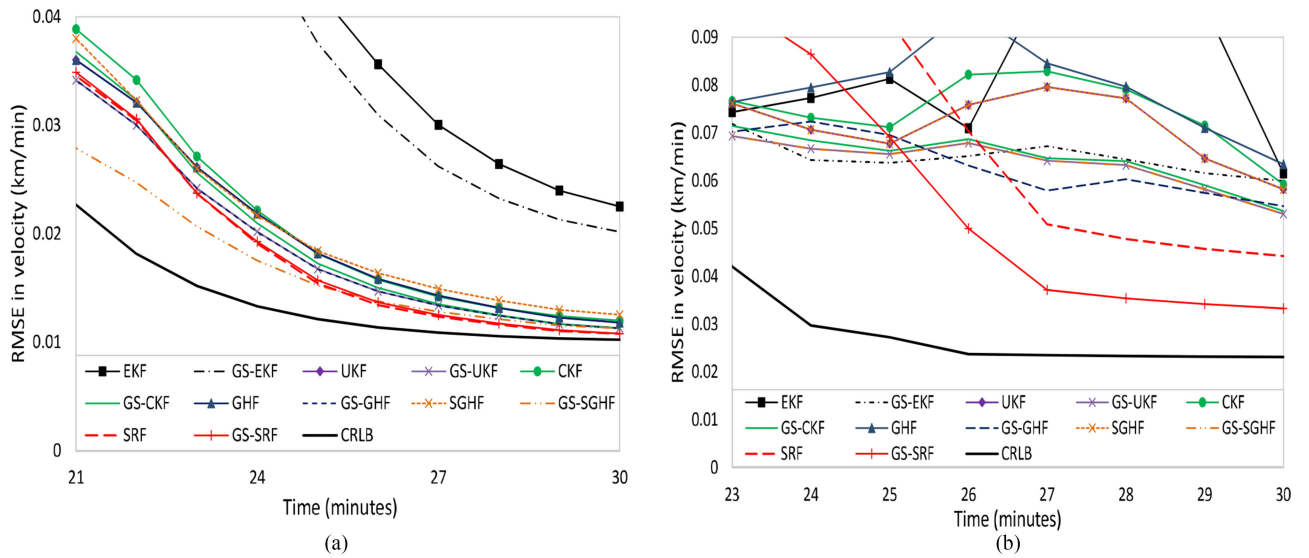


Fig. 4. RMSE in velocity for (a) case 1 and (b) case 2.

for case 2. The effect of increasing the measurement noise in RMSE for all the filters is studied by plotting $RMSE_{k_{max}}$ for varying σ_θ for both cases, where Fig. 6(a) is for case 1 and Fig. 6(b) is for case 2. From the figures, it can be noted that an increase in the measurement noise caused an increase in RMSE at the final step. However, it can be observed that the performance of GS-SRF is far better than all the filters, for both cases.

The tracking efficiency of GS-SRF in terms of track divergence is shown in Table II for case 1, where the effect of initial covariance, measurement noise covariance, and sampling time on track divergence is studied. Here, the value of T_0 is set to be 1 km. It can be noted that the number of diverged tracks steadily increased with an increase in μ for all the quadrature filters. For the same case, SRF and GS-SRF performed with superior tracking accuracy with not even a single diverged track. Table II also gives results of % track divergence when the measurement

noise σ_θ is varied. From these results, we can note that the tracking performance of quadrature filters suffered when the measurements were highly noise corrupted. However, GS-SRF performed with superior accuracy with zero number of track divergence, whereas for $\sigma_\theta = 4^\circ$, SRF incurred a single diverged track.

An increase in sampling time resulted in a decrease in accuracy for all the filters except SRF and GS-SRF, as shown in Table II. All the quadrature filters incurred higher track divergence, whereas SRF and GS-SRF did not even loose a single track. From the results in Table II, it can be observed that out of all the Gaussian filters, EKF gave the worst filtering performance while other quadrature-based filters performed with comparable accuracy. On the other hand, newly formulated GS-SGHF gave comparable results with other GS filters. The main highlight of the results presented in Table II is the filtering accuracy of SRF and the proposed GS-SRF. It can be observed that GS-SRF performed with

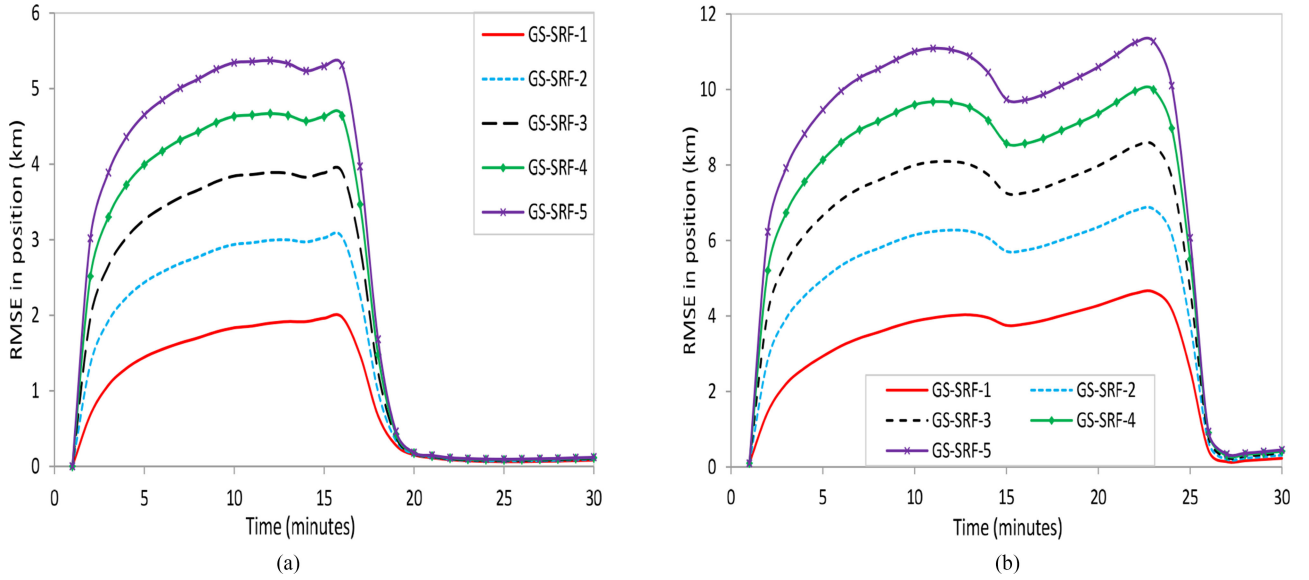


Fig. 5. RMSE in position for various μ for (a) case 1 and (b) case 2.

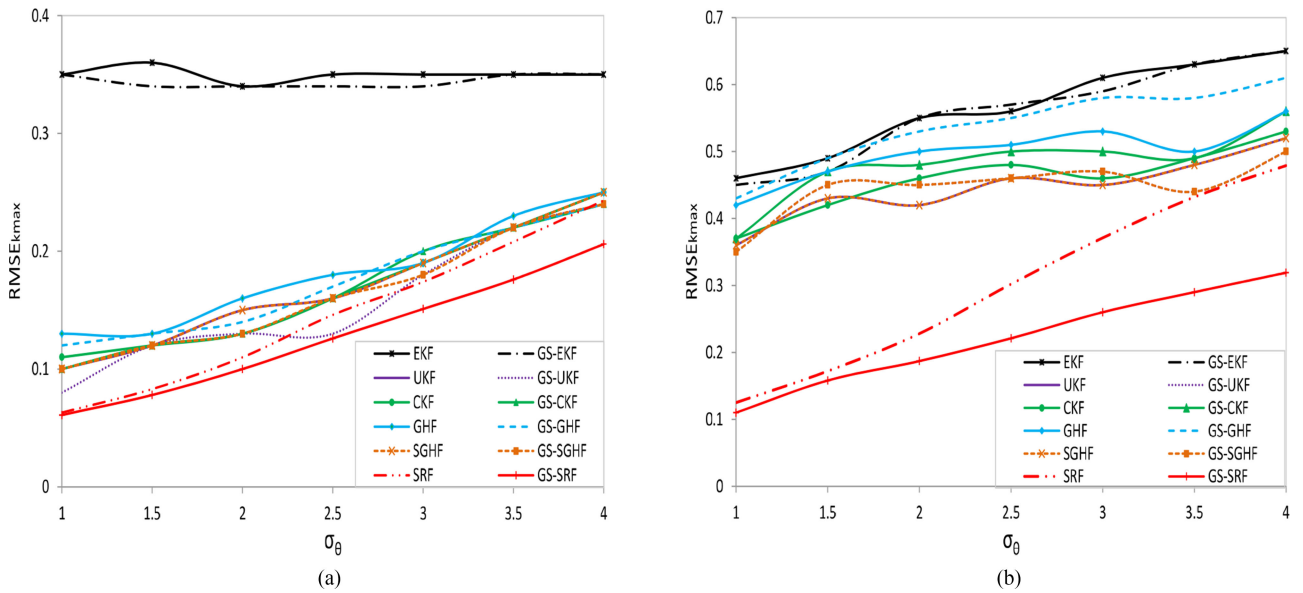


Fig. 6. $RMSE_{kmax}$ for various σ_θ for (a) case 1 and (b) case 2.

zero track divergence irrespective of an increase in initial uncertainty, measurement noise, and sampling time. Relative computational time for all the filters is listed in Table II. From this, it can be noted that computational load of GS-SRF is comparable to other GS quadrature filters.

Table III shows the superior filtering accuracy of GS-SRF, where case 2 is considered. Here also, the value of T_b is set to 1 km. An increase in initial covariance resulted in an almost complete track divergence for all the quadrature filters, whereas GS-SRF suffered from a maximum track divergence of only 5%, which is less than the SRF track divergence. The effect of an increase in measurement noise also resulted in a very high track divergence for quadrature filters and their GS extensions. It can be noted that for small values of σ_θ , SRF and GS-SRF incurred almost the same number of diverged tracks. However, for higher values, the filtering performance of GS-SRF is more accurate such that the number

of diverged tracks was lower when compared to SRF. Similar kind of performance was observed for all the filters when sampling time was increased, as mentioned in Table III. From all these results, it can be noted that except for SRF and GS-SRF, all other Gaussian and GS filters failed with unacceptable track divergence percentage. Out of SRF and the proposed GS-SRF, the latter performed with better track accuracy.

To further investigate the filtering accuracy of the proposed filters, the track divergence condition was assumed to be a function of standard deviation of computed position error distribution. It was found that the average standard deviation of position error for case 1 was approximately equal to 100 m. For $T_b = 100$ m track divergence condition, the proposed GS-SGHF and GS-SRF incurred a percentage track divergence of 26.79% and 23.47%, respectively, while SGHF and SRF incurred 28% and 26%, respectively. Now setting the track

TABLE II
% OF TRACK DIVERGENCE FOR VARIOUS FILTERS WITH RESPECT TO μ , σ_θ , AND T FOR CASE 1

		EKF	UKF	CKF	GHF	SGHF	GS-EKF	GS-UKF	GS-CKF	GS-GHF	GS-SGHF	SRF	GS-SRF
μ	1	0.84	0.54	0.58	0.41	0.53	0.25	0.12	0.09	0.1	0.12	0	0
	2	1.43	0.71	0.67	0.55	0.71	0.25	0.13	0.11	0.13	0.12	0	0
	3	2.2	0.94	0.95	0.69	0.92	0.28	0.17	0.2	0.16	0.17	0	0
	4	2.87	1.22	1.26	0.96	1.22	0.34	0.21	0.28	0.19	0.2	0	0
	5	3.64	1.52	1.47	1.06	1.52	0.4	0.29	0.31	0.26	0.27	0	0
σ_θ	1	0.54	0.33	0.15	0.23	0.33	0.23	0.09	0.02	0.08	0.09	0	0
	1.5	0.84	0.54	0.58	0.41	0.53	0.25	0.12	0.09	0.1	0.12	0	0
	2	2.51	1.28	1.12	0.98	1.28	0.63	0.39	0.26	0.35	0.36	0	0
	2.5	3.04	1.59	1.11	1.22	1.59	0.78	0.43	0.31	0.45	0.43	0	0
	3	3.82	1.68	1.51	1.52	1.68	0.92	0.54	0.35	0.49	0.54	0	0
	3.5	4.42	1.89	1.73	1.76	1.89	1.22	0.57	0.39	0.53	0.55	0	0
	4	5.36	2.42	2.24	2.2	2.4	1.36	0.58	0.54	0.57	0.59	0.01	0
T	1	0.84	0.54	0.58	0.41	0.53	0.25	0.12	0.09	0.1	0.12	0	0
	1.5	7.57	1.4	1.5	2.2	1.3	7.07	0.8	0.7	0.9	0.6	0	0
	2	29.3	2.6	2.2	5.1	2.4	27.8	1.8	1.7	3.2	1.9	0	0
R. c. time		1	2.1	2.1	6.6	2.2	1.5	2.9	2.9	15.7	3.2	2.1	3.3

TABLE III
% OF TRACK DIVERGENCE FOR VARIOUS FILTERS WITH RESPECT TO μ , σ_θ , AND T FOR CASE 2

		EKF	UKF	CKF	GHF	SGHF	GS-EKF	GS-UKF	GS-CKF	GS-GHF	GS-SGHF	SRF	GS-SRF
μ	1	77.2	72	75.6	74.2	72.3	73.7	70.5	73.2	75.9	73.6	1	0.8
	2	85.6	83.8	82.7	82.3	83.2	84.2	81.6	80.9	80.2	81.7	1.4	1.1
	3	88.4	86.5	85.8	85.3	86	87.9	82.8	82.4	82.1	82.4	4	3.4
	4	90.1	89.2	89	88.7	89.8	89.2	86.7	86.2	86	86.5	5.7	4.3
	5	92.9	91.7	90.1	89.5	91.4	91.6	90	89.4	88.1	89.6	7.9	4.8
σ_θ	1	56.5	37.1	36.4	35.5	37.3	54.2	33.1	33.9	37.3	33.1	0.3	0.1
	1.5	68.1	60.1	58.3	59.2	59.7	67.4	56.9	55.8	58.7	56.4	0.5	0.5
	2	77.2	72	75.6	74.2	72.3	73.7	70.5	73.2	75.9	73.6	1	0.8
	2.5	82.9	79.8	81.1	78.2	80.2	77.3	75.8	76.1	75.1	75.8	2	1.3
	3	84.2	83.9	82.4	83.1	83.5	80.4	79.1	80.2	79.4	79.9	2.5	2.2
	3.5	87.5	85.1	84.7	83.9	84.7	83.8	81.7	82.1	81.3	81.8	4.9	2.8
	4	89.6	87.8	87.4	88.2	89.2	88.2	86.2	85.4	87.7	86.5	7.8	3.4
T	1	77.2	72	75.6	74.2	72.3	73.7	70.5	73.2	75.9	73.6	1	0.8
	1.5	82	79.3	76.2	80.4	79.1	80.2	79.8	76.6	81.6	75.1	3.5	1.2
	2	84.8	81.2	81.2	81.9	81.4	81.8	80.8	78.1	82.8	79.2	6.4	3.9

divergence condition to 300 m, the proposed GS-SGHF and GS-SRF incurred a percentage track divergence of 0.18% and 0.02%, respectively. The corresponding track divergence for SGHF and SRF was 0.56% and 0.11%, respectively. For a 500-m track divergence condition, SRF and GS-SRF performed without even a single diverged track, whereas SGHF and GS-SGHF performed with 0.54% and 0.14% track divergence, respectively.

For case 2, the average standard deviation of position error was found to be approximately equal to 500 m. Now setting the track divergence condition to $T_b = 500$ m, the percentage of track divergence incurred by SRF and GS-SRF was 5.1% and 3.5%, respectively. For this condition, SGHF and GS-SGHF incurred an almost complete track divergence. To further explore the tracking efficiency of GS-SRF, track divergence condition was set to 300 m. In this case, GS-SRF incurred a percentage track divergence of 14.2% while SRF incurred 24.4%. This again proves the superior accuracy of GS-SRF compared to SRF.

For examining the consistency of the proposed filters, NEES was calculated for both cases 1 and 2. The chi-square test was performed by defining a 95% probability concentration region, that is, $\alpha = 0.05$ in (25). The hypothesis test was conducted over 1000 Monte Carlo runs, resulting in a concentration region with lower and upper limits $r_l = 3.82$ and $r_u = 4.17$.

Fig. 7(a) and (b) shows the average NEES for cases 1 and 2, respectively. From Fig. 7(a), it can be noted that the average NEESs of the proposed GS-SRF and the SRF lie well within the probability concentration region, which is the criterion for a consistent filter. Since EKF and GS-EKF incurred very high error values, their NEES also incurred very high values and is not shown in the figure. From the rest of the filters implemented, average NEESs of the proposed GS-SGHF and other filters such as GHF, GS-GHF, GS-UKF, and GS-CKF lie just below r_l , proving the consistency of these filters. It was observed that all GS quadrature filters proved to be consistent in terms of NEESs. Fig. 7(b) shows the average NEESs for all the filters except EKF and GS-EKF (for the reasons mentioned above) for case 2. In this case, except for GS-SRF, all other filters incurred very high average NEES values well above the probability concentration region. Hence, from this consistency study, it can be inferred that GS-SRF proved to be the only consistent filter among all the filters implemented (for both cases).

VI. DISCUSSION AND CONCLUSION

In this paper, we develop GS-SRF and GS-SGHF for an underwater bearings-only target tracking problem. To make a fair comparative study, GS-CKF, GS-GHF, GS-UKF, and GS-EKF are also

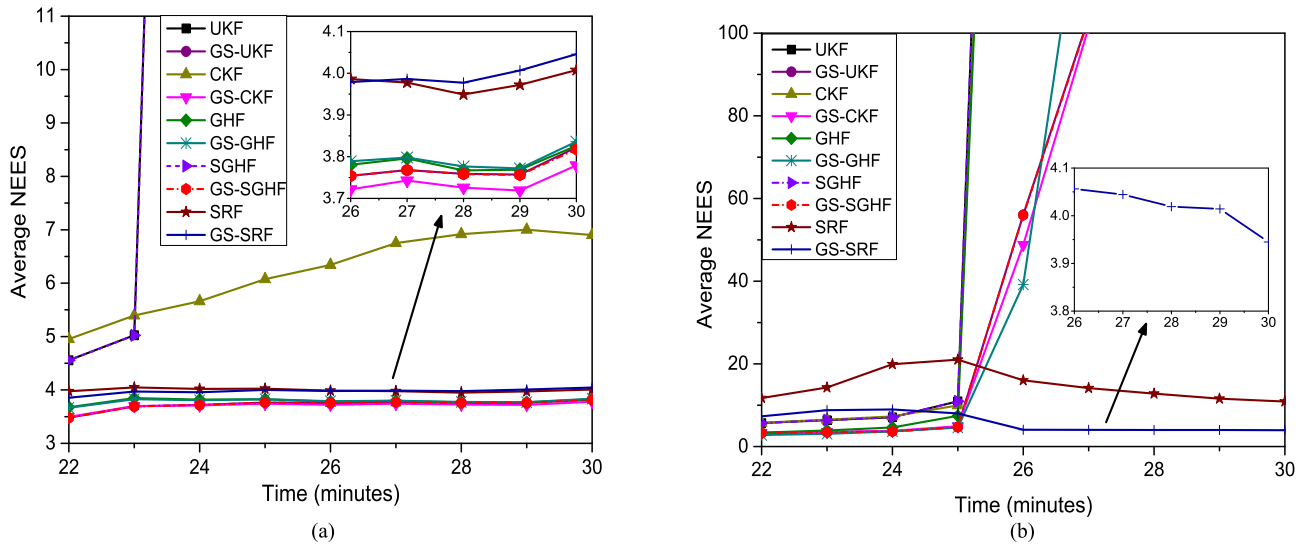


Fig. 7. Average NEESs for (a) case 1 and (b) case 2.

implemented. Performance of the filters was studied by plotting the RMSE and comparing it with that of CRLB. Other parameters considered were the percentage of track divergence and the computational time. The effect on tracking accuracy due to initial uncertainties, measurement noise covariance, and sampling time is also studied. The findings of this paper can be summarized as follows.

- 1) Out of all the filters, GS-SRF gives the overall best result with better accuracy, less track divergence, and moderate computational time.
- 2) Increase in initial uncertainty resulted in an increase in track divergence count for all the filters. GS-SRF proved to be more robust and accurate than all other filters.
- 3) Increasing the measurement noise resulted in lowering the filtering accuracy. RMSE at last instant for all the filters was increased, and out of these, GS-SRF gave more accurate results.
- 4) Varying the sampling time had a considerable effect on the number of diverged tracks. GS-SRF gave the best result in this case with a lower track divergence percentage.
- 5) Consistency of the filters was studied by calculating the average NEES. From this study, it can be concluded that the proposed GS-SRF proved to be the only consistent filter for both cases combined.

To conclude, the proposed GS-SRF proves to be an efficient tracking algorithm for the BOT problem with good tracking accuracy and moderate computational load. Hence, GS-SRF can be considered as a better solution for the BOT scenarios considered.

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Rahul Radhakrishnan received the B.Tech. degree in applied electronics and instrumentation from the Government Engineering College, Calicut, India, in 2010 and the M.Tech. degree in control systems from the Department of Electrical Engineering, National Institute of Technology Kurukshetra, India, in 2013. He is currently working toward the Ph.D. degree at the Department of Electrical Engineering, Indian Institute of Technology Patna, India.

His main research interests include nonlinear filtering, aerospace, and underwater target tracking.



Shovan Bhaumik received the B.Sc. degree in physics from the University of Calcutta, Kolkata, India, in 1999 and the B.Tech. degree in electronics and instrumentation engineering and the M.E. and Ph.D. degrees in electrical engineering from Jadavpur University, Kolkata, India, in 2002, 2004, and 2009, respectively.

From 2007 to 2009, he was a Research Engineer with GE Global Research, GE India Technology Center, Bangalore, India. He is currently an Associate Professor at the Department of Electrical Engineering, Indian Institute of Technology Patna, Patna, India. His main research interests include nonlinear estimation, statistical signal processing, and underwater target tracking.



Nutan Kumar Tomar received the Ph.D. degree in exact and approximate controllability of semilinear control systems from the Department of Mathematics, Indian Institute of Technology Roorkee, Roorkee, India, in 2008.

He is an Associate Professor at the Department of Mathematics, Indian Institute of Technology Patna, Patna, India. His main research interests include the controllability and observability problems of control systems.